

Akshay Kumar

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Summary

Motivated and enthusiastic Data Science fresher with strong analytical skills. Proficient in **Python** with hands-on experience in Machine Learning, Deep Learning, and Data Visualization. Seeking to apply academic knowledge to solve real-world problems in a data-driven environment.

Experience

Data Science Intern

Feb 2024 – June 2024

National Institute of Technology, Kurukshetra

- Led data wrangling and feature engineering for predictive modeling projects.
- Applied machine learning and deep learning models for trend analysis and sentiment extraction using NLP.
- Optimized models through hyperparameter tuning to improve performance.
- Technologies used: **Python**, **TensorFlow**, **Scikit-learn**, NLP techniques.

Projects

Stock Trend Prediction Using LSTM Neural Networks

- Developed a time-series forecasting model to predict future stock prices using Long Short-Term Memory (LSTM) neural networks.
- Preprocessed historical stock price data (closing prices) and trained the LSTM model to capture long-term dependencies.
- Achieved significant performance improvement in trend prediction through hyperparameter tuning and model optimization.
- Visualized predicted vs actual stock prices using Matplotlib to demonstrate the model's effectiveness.
- **Technologies used:** Python, TensorFlow, Keras, Pandas, NumPy, Matplotlib.
- **GitHub Repository:** [Link](#)

PDF Extraction with AI Integration

Date

Developed a Flask-based web application for extracting structured data from PDF invoices using AI. The system processes invoices by extracting text from the uploaded PDFs and uses LangChain and OpenAI's GPT-3.5-turbo model to automatically extract relevant information like export codes, weight, and transport details.

- Built a Flask web application for uploading and processing PDF invoices.
- Integrated OpenAI GPT-3.5-turbo model with LangChain and Kor for extracting data.
- Defined a custom schema using the 'kor.nodes' library to extract key attributes from invoices (e.g., document type, codes, weights).
- Enabled automatic extraction of invoice details such as GSTIN, PAN, and other relevant fields.
- Technologies used: **Flask**, **PyPDF2**, **LangChain**, **OpenAI GPT-3.5**, **Kor**, **Python**.
- GitHub Repository: [Link](#).

Education

Master of Science in Computer Science

2022 – 2024

Central University of Rajasthan

- Specialization in Big Data Analytics.

Bachelor of Computer Applications

2019 – 2022

Raj Narayan College Hajipur, Babasaheb Bhimrao Ambedkar Bihar University

- Major in Web Design and Development.

Skills

- **Programming Languages:** Python **Web Development:** Django Framework **Databases:** SQL, NoSQL (MySQL)
- **Data Analysis & Visualization:** Pandas, NumPy, EDA, Power BI, Excel, Seaborn, Matplotlib, Plotly, Bokeh
- **Machine Learning & Artificial intelligence:** Regression, Classification, Clustering, NLP, Gen AI, Langchain, Hugging Face. **Tools:** VS Code, Git, Jupyter
- **Computer Vision:** OpenCV, Image Processing
- **Big Data Technologies:** PySpark, Apache Hive **Mathematics:** Linear Algebra, Statistics

Certifications

- **Python for Data Science – Cognitive Class** **Machine Learning Specialization – Cognitive Class**

Hobbies

- Chess (1st rank in CURAJ Chess Competition) Programming challenges